

Unit 5, Lesson 1: Exploring Side-Side-Side Triangle Congruence



Benchmarks

MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle, and Hypotenuse-Leg.

MA.912.GR.2.1 Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates.

MA.912.GR.5.1 Construct a copy of a segment or an angle.



Learning Target

- ☐ I can use constructions and transformations to determine if SSS congruence can be used to show that two triangles are congruent.

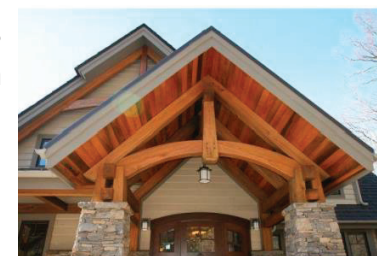


5.1.1 Warm-Up: STEM

Triangles are effective tools for architecture used in the design of buildings and other structures, as they provide strength and stability.

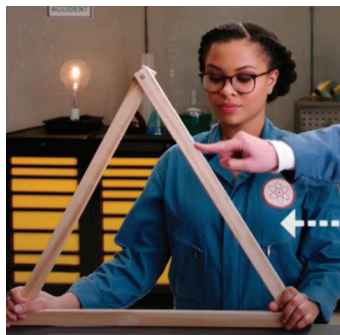


When building materials are used to form a triangle, the design has a heavy base and the pinnacle on the top is capable of handling weight because of how the energy is distributed throughout the triangle. The triangle's use in architecture dates back longer than other common architectural shapes such as the dome, arch, and cylinder. It even predates the wheel.



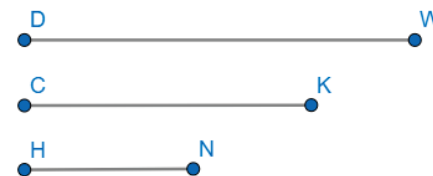
1. Imagine using straws or toothpicks to make a square or rectangle. What do you think will happen when the engineer puts pressure on the side of the rectangle?

2. Now imagine you made a triangle using straws or toothpicks. What do you think will happen when the engineer puts pressure on the side of the triangle?



5.1.2 Exploration: Are Three Sides Enough?

1. $\overline{DW}$, $\overline{CK}$, and $\overline{HN}$ are given. Use the given steps to construct $\triangle BYT$ so that its sides are congruent to $\overline{DW}$, $\overline{CK}$, and $\overline{HN}$.



Step 1: Copy $\overline{DW}$. Construct $\overline{BY}$ so that it is congruent to $\overline{DW}$.

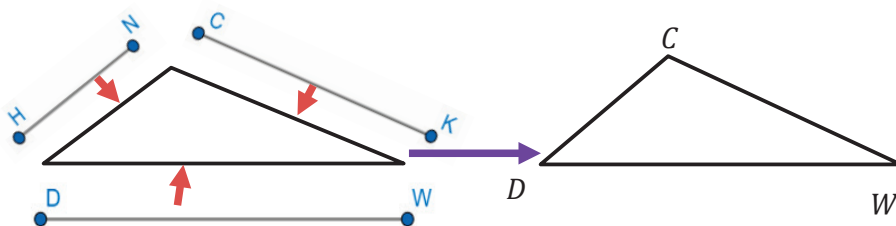
Step 2: Copy $\overline{CK}$. Place the compass on point B to create an arc that is the length of $\overline{CK}$.

Step 3: Copy $\overline{HN}$. Place the compass on point Y to create an arc that is the length of $\overline{HN}$.

Step 4: Mark the intersection of the two arcs using a point. Label the point T .

Step 5: Draw $\overline{BT}$ and $\overline{YT}$.

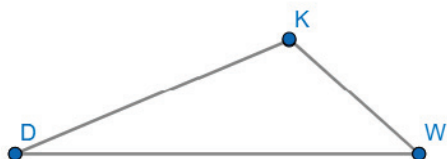
2. The triangle below was created using the three given line segments.



Compare the triangle you constructed with $\triangle CDW$. Describe any similarities or differences.

3. Maria used a computer program to create line segments $\overline{DW}$, $\overline{CK}$, and $\overline{HN}$. She then moved the line segments to construct $\triangle DKW$. Because the vertices of the line segments $\overline{CK}$ and $\overline{HN}$ coincided with other vertices, she only used the letters D , K , and W .

- a. What do you notice about $\triangle BYT$ and $\triangle DKW$?

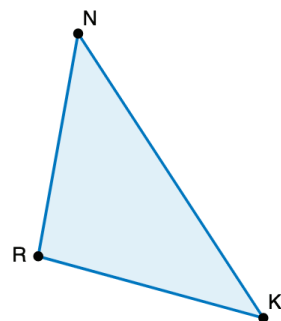
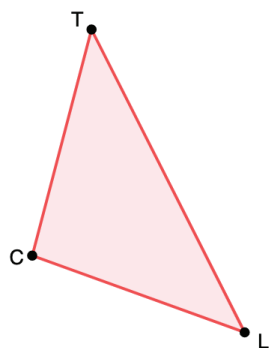


- b. The line segments $\overline{DW}$, $\overline{CK}$, and $\overline{HN}$ were used to construct three different triangles. Work with your partner to write a conjecture about creating triangles whose sides are congruent.

- c. Ask two classmates about their conjectures. Record their conjectures and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Conjecture for Creating Triangles Whose Sides are Congruent	My Summary of Their Reason	Initials

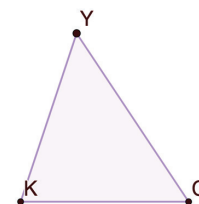
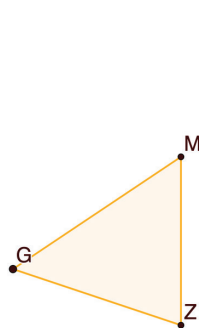
4. Use patty paper to determine if $\triangle CTL$ and $\triangle RNK$ were created using the same three line segments.



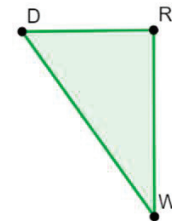
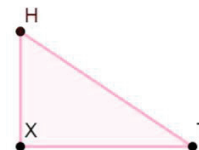
5. Discuss the following with your partner: If three sides of a triangle are congruent, does that mean all of the angles of the triangles are congruent too? Explain your decision and state the evidence used for the decision.

6. Two sets of triangles are shown.

$\triangle MGZ$ and $\triangle KYC$

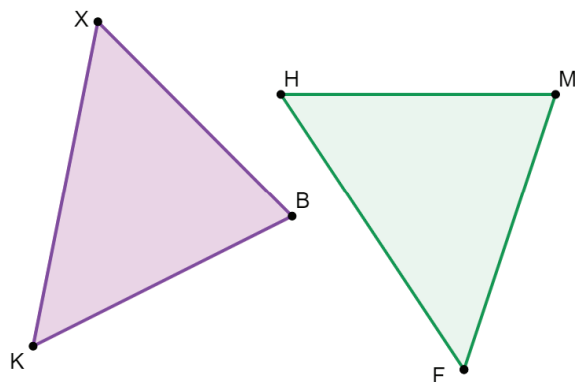


$\triangle HTX$ and $\triangle WDR$



- Use patty paper to determine if the sides of the two triangles are congruent.
- $\triangle$ _____ and $\triangle$ _____ are congruent because all of the sides are congruent.
- For the two triangles that are congruent, use patty paper to verify that the angles are congruent.
- For the two triangles that are congruent, list the sides on each triangle that are congruent to each other.

7. $\triangle KXB$ and $\triangle HFM$ are shown. $HM = 4.2$, $XB = 4.2$, $XK = 5$, and $KB = 4.4$.



- a. Discuss with your partner if the given measurements are sufficient to determine whether $\triangle KXB$ and $\triangle HFM$ are congruent. Explain your decision.
- b. Use patty paper, a protractor, a ruler, or a compass to measure all of the parts of $\triangle KXB$ and $\triangle HFM$. Then, complete the sentence.

$\triangle KXB$ and $\triangle HFM$ are

- ☐ congruent
☐ not congruent

because

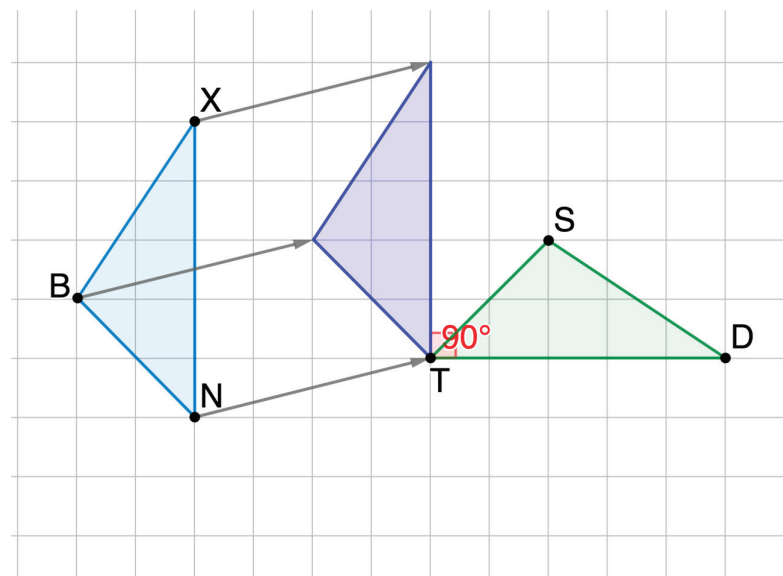
side(s) and

- ☐ 0
☐ 1
☐ 2
☐ 3

angle(s) are congruent.

- ☐ 0
☐ 1
☐ 2
☐ 3

8. A series of transformations shows how $\overline{BX}$, $\overline{XN}$, and $\overline{BN}$ were translated, then rotated 90° about point T to create $\overline{SD}$, $\overline{DT}$, and $\overline{TS}$.



- a. Use the grid to determine if $BN = TS$, $BX = DS$, and $XN = TD$.
- b. Discuss with your partner how the transformations that were used ensure that the length of each line segment was preserved.

- c. Use the word bank to complete the statements. Words may be used more than once.

Word Bank				
angles	congruent	congruence	parts	rigid motions

Since $\overline{BX}$, $\overline{XN}$, and $\overline{BN}$ were translated then rotated 90° , the length of each line segment was preserved because of the definition of congruence in terms of _____. The definition of congruence in terms of _____ also ensures that the _____ are congruent. Because all of the _____ of $\triangle XNB$ are _____ to all of the corresponding _____ of $\triangle DTS$, the two triangles are _____.



5.1.3 Bring It Together: Triangle Congruence



Side-Side-Side (SSS) Congruence Postulate

Two triangles are congruent if the three sides of one are congruent to the three sides of the other, respectively.

1. Work with your partner to explain how the exploration has shown that the Side-Side-Side (SSS) Congruence Postulate is true.



5.1.4 Wrap-Up: Cool Down

1. Maggie and Naomi both constructed a triangle using the side lengths shown.

- $CN = 2.5$ inches (in.)
- $NK = 3.75$ in.
- $CK = 5.25$ in.

Explain whether it is possible to find out if the two triangles Maggie and Naomi created will be congruent.



Unit 5, Lesson 2: Exploring Side-Angle-Side and Angle-Side-Angle Triangle Congruence



Benchmarks

MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle, and Hypotenuse-Leg.

MA.912.GR.5.1 Construct a copy of a segment or an angle.



Learning Targets

- ☐ I can use constructions to determine if SAS congruence can be used to show that two triangles are congruent.
- ☐ I can use constructions to determine if ASA congruence can be used to show that two triangles are congruent.
- ☐ I can use constructions to determine if SSA congruence can be used to show that two triangles are congruent.